

County-Level MSM Population Estimation

We estimate the population of men who have sex with men (MSM) at the county level using a multi-step approach that incorporates household structure, urbanicity stratification, and demographic weighting. The method builds on prior work by Grey et al. and Oster et al., with a slight adjustment in methods and sources and extending the same algorithm for prediction of demographic subgroup populations by Age also to Income, Race, and Education.

1. Estimation of Same-Sex Male Households

We assume that the ratio of male-male (MM) to total same-sex (MM + FF) households is consistent across counties within a given urbanicity and state. To operationalize this, we compute the following proportion for each state s and urbanicity u :

$$\rho_{s,u} = \frac{\sum_{c \in s,u} \text{MM}_c}{\sum_{c \in s,u} (\text{MM}_c + \text{FF}_c)} \quad (1)$$

where:

- MM_c : Number of male-male same-sex households in county c
- FF_c : Number of female-female same-sex households in county c

We then use this value to estimate the number of same-sex male (SSM) households in each county as:

$$\widehat{\text{SSM}}_c = \rho_{s,u} \cdot \text{SS}_c \quad (2)$$

where SS_c is the total number of same-sex households in county c .

Assumption: This adjustment assumes that the $\text{MM}/(\text{MM}+\text{FF})$ composition is stable within each state-urbanicity pairing and reflects meaningful differences in couple distributions by region.

Imputation for Missing SSM Household Counts In cases where $\widehat{\text{SSM}}_c$ is missing due to zero-suppression or data unavailability, we impute values using urbanicity-stratified averages. Specifically, we:

Add an imputed value based on the average SSM household rate for the urbanicity stratum:

$$\widehat{\text{SSM}}_c \leftarrow \widehat{\text{SSM}}_c + \left(\frac{\sum_{c \in u} \widehat{\text{SSM}}_c}{\sum_{c \in u} \text{HH}_c} \right) \cdot \text{HH}_c$$

where u denotes the urbanicity group and HH_c is the total number of households in county c .

This ensures consistency in household-based MSM estimation across counties with suppressed or incomplete SSM data.

2. MSM Household Index

We construct a household-based index to scale county-level estimates relative to urbanicity-wide norms:

$$\text{Index}_c = \frac{\widehat{\text{SSM}}_c / \text{HH}_c}{\overline{\text{SSM}}_u / \overline{\text{HH}}_u} \quad (3)$$

where:

- HH_c : Total households in county c
- $\overline{\text{SSM}}_u, \overline{\text{HH}}_u$: Aggregated SSM households and total households in urbanicity u

3. Base MSM Proportion by Urbanicity

From the General Social Survey (GSS), we extract urbanicity-stratified base MSM proportions:

$$P_u = \mathbb{P}(\text{MSM} \mid u) \quad (4)$$

4. Initial County-Level MSM Estimate

We estimate the MSM population in county c as:

$$\widehat{M}_c = P_u \cdot \text{MalePop}_c \cdot \text{Index}_c \quad (5)$$

where MalePop_c is the number of adult males in county c , taken from the ACS.

5. Demographic Breakdown

To distribute county-level MSM estimates into demographic subgroups (e.g., by age, education, income), we use urbanicity-specific subgroup proportions from the GSS:

1. Compute unadjusted subgroup estimates:

$$\widehat{M}_{c,d} = p_{u,d} \cdot \text{MalePop}_{c,d}$$

2. Normalize subgroup shares within county:

$$s_{c,d} = \frac{\widehat{M}_{c,d}}{\sum_{d'} \widehat{M}_{c,d'}}$$

3. Scale to match county total MSM:

$$\widehat{M}_{c,d} = s_{c,d} \cdot \widehat{M}_c$$

6. Simulation Method

Quantity Simulated	Distribution Used
Male population (county-level)	Normal (mean from ACS, SE from MOE/1.645)
Male population (age, education, income groups)	Normal (group-specific estimates, with SE from MOE)
Total households (county-level)	Normal (with SE from MOE)
Same-sex households (MM, FF, SS)	Log-normal (after log-transform of mean and SE)
MSM proportion (by urbanicity)	Logit-normal
MSM proportion (by demo $\times$ urbanicity)	Logit-normal
MM / (MM + FF) ratio	Logit-normal (logit transformed ratio, SE derived from counts)

Table 1: Simulated input quantities and assumed distributions for uncertainty propagation

We use the following transformation to simulate draws from a logit-normal distribution:

$$\mu = \text{logit}(p), \quad \text{SE}_{\text{logit}} = \frac{\text{SE}}{p(1-p)}, \quad p^{(\text{draw})} = \text{logit}^{-1}(\mathcal{N}(\mu, \text{SE}_{\text{logit}}))$$

Any simulated values that resulted in negative estimates (e.g., due to draws from the normal distribution) were set to zero to preserve interpretability and validity.

7. Data Sources

The data sources are detailed below.

Data Source	Variable(s)	Usage in MSM Estimation
ACS 2022	Total households (HH_c) Male population 18+ ($MalePop_c$)	Denominator for SSM index Base population for MSM count calculation
ACS 2023 (Age, Educ), 2021 (Income)	Demographic distributions: age, education, income	Weighting male population by subgroup for demographic breakdown
ACS 2014	Same-sex male households (MM_c), Female same-sex (FF_c), Total same-sex (SS_c)	Used to compute MM/SS ratio Derivation of SSM index
ACS 2022 (for SS)	Same-sex households (SS)	Used to update total SS counts by county
GSS 2016–2024	MSM prevalence by urbanicity (P_u) MSM prevalence by subgroup and urbanicity ($p_{u,d}$)	Determines baseline and demographic-specific MSM rates
NCHS Urban-Rural Classification	Urbanicity designation (u) for each county	Stratifies counties for applying appropriate MSM proportions

Table 2: Data sources, variables, and their roles in MSM population estimation

Details on Data Cleaning, Harmonization, and Assumptions

Prior to estimation, we applied several steps to clean and harmonize the demographic and behavioral data used in the MSM estimation pipeline.

GSS Data Cleaning

- **Years Included:** We used GSS years 2016, 2018, 2020, 2021, 2022, 2024 to maximize power and reduce uncertainty. Years before 2016 were excluded due to missing income variable `income16`.
- **MSM Definition:** Respondents were classified as MSM if they were male and reported male or male-and-female sexual partners in the past 12 months.
- **Weighting:** `wtssnrps` was used to appropriately weight individuals according to GSS survey design. These are described below:

- **Urbanicity Classification:** Using GSS variable `xnorcsiz`, counties were mapped to urbanicity types as follows:
 1. **Large central metropolitan county:** a large central city (population over 250,000)
 2. **Large fringe metropolitan county:** a suburb or unincorporated area of a large city
 3. **Medium/small metropolitan county:** medium cities (50,000–250,000) and smaller towns (10,000–49,999)
 4. **Nonmetropolitan county:** towns and rural areas with populations under 10,000, including unincorporated regions
- **Variable Filtering:** Respondents with missing values in gender, urbanicity, or MSM status variables were excluded.

ACS Data Cleaning

- **Male Population:** We filtered to males aged 18+ for age and race breakdowns. For education and income, total population estimates were used, assuming distributions are similar across sexes.
- **Demographic Harmonization:** ACS and GSS variables were grouped to align:
 - **Age:** Derived from the variable `age`, grouped into:
 - * 18–24
 - * 25–34
 - * 35–44
 - * 45–54
 - * 55+
 - **Education:** Derived from the variable `educ`, grouped into:
 - * Less than high school (e.g., 0–11 years)
 - * High school graduate (12 years)
 - * Some college, including associate’s degrees (13–15 years)
 - * Bachelor’s degree or higher (16+ years)
 - **Household Income:** Based on the variable `income16`, grouped into:
 - * Less than \$20,000
 - * \$20,000 to \$40,000
 - * \$40,000 to \$100,000
 - * \$100,000 or more
 - **Household Income:** Based on the variable `race`, grouped into:
 - * White

- * Black
- * Other
- * Hispanic